

Experiment No.9: Process Explorer

Aim

To use Process Explorer to examine running processes, analyze CPU and memory usage, inspect process properties, check network activity, and identify potentially suspicious processes.

Software Required

- Windows Operating System
- Process Explorer (Sysinternals)
- Windows Security
- Web Browser

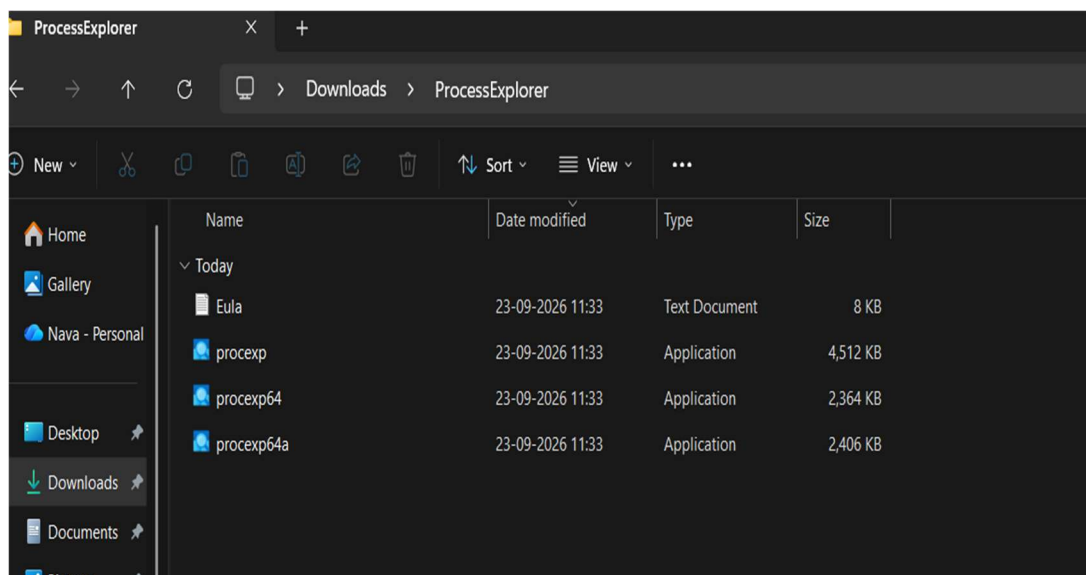
Procedure

Step 1 – Download and Extract Process Explorer

Process Explorer was downloaded and extracted into the `ProcessExplorer` folder.

The folder contained:

- procexp.exe
- procexp64.exe
- procexp64a.exe
- Eula.txt



Step 2 – Run Process Explorer as Administrator

The `procxp64.exe` application was executed using ****Run as administrator****.

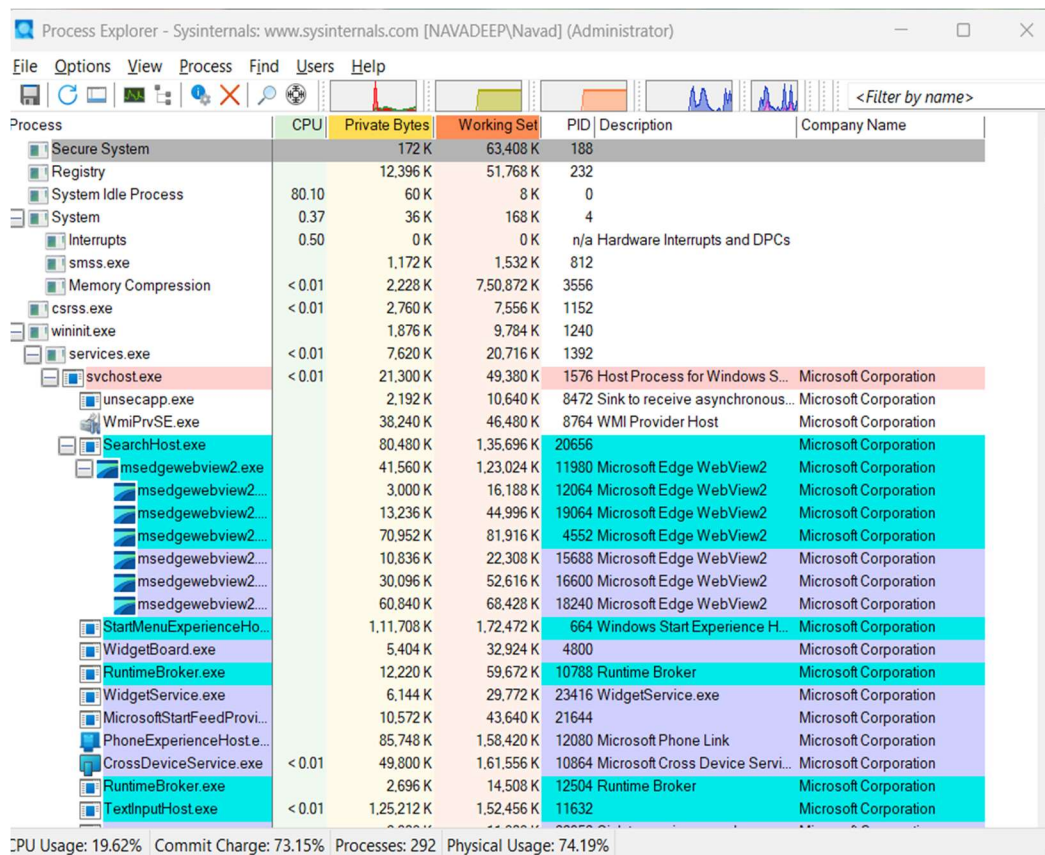
Process Explorer displayed the running processes in a hierarchical process tree.

Step 3 – Examine Running Processes

The Process Explorer window was examined using the following columns:

- Process
- CPU
- Private Bytes
- Working Set
- PID
- Description
- Company Name

The process tree was inspected to identify processes and their resource usage.



The screenshot shows the Process Explorer window with the following columns: Process, CPU, Private Bytes, Working Set, PID, Description, and Company Name. The process tree on the left shows the hierarchy starting from System, with various system processes and user applications. The table on the right lists the following processes:

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		172 K	63,408 K	188		
Registry		12,396 K	51,768 K	232		
System Idle Process	80.10	60 K	8 K	0		
System	0.37	36 K	168 K	4		
Interrupts	0.50	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,172 K	1,532 K	812		
Memory Compression	< 0.01	2,228 K	7,50,872 K	3556		
csrss.exe	< 0.01	2,760 K	7,556 K	1152		
wininit.exe		1,876 K	9,784 K	1240		
services.exe	< 0.01	7,620 K	20,716 K	1392		
svchost.exe	< 0.01	21,300 K	49,380 K	1576	Host Process for Windows S...	Microsoft Corporation
unsecapp.exe		2,192 K	10,640 K	8472	Sink to receive asynchronous...	Microsoft Corporation
WmiPrvSE.exe		38,240 K	46,480 K	8764	WMI Provider Host	Microsoft Corporation
SearchHost.exe		80,480 K	1,35,696 K	20656		Microsoft Corporation
msedgewebview2.exe		41,560 K	1,23,024 K	11980	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		3,000 K	16,188 K	12064	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		13,236 K	44,996 K	19064	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		70,952 K	81,916 K	4552	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		10,836 K	22,308 K	15688	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		30,096 K	52,616 K	16600	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		60,840 K	68,428 K	18240	Microsoft Edge WebView2	Microsoft Corporation
StartMenuExperienceHo...		1,11,708 K	1,72,472 K	664	Windows Start Experience H...	Microsoft Corporation
WidgetBoard.exe		5,404 K	32,924 K	4800		Microsoft Corporation
RuntimeBroker.exe		12,220 K	59,672 K	10788	Runtime Broker	Microsoft Corporation
WidgetService.exe		6,144 K	29,772 K	23416	WidgetService.exe	Microsoft Corporation
MicrosoftStartFeedProvi...		10,572 K	43,640 K	21644		Microsoft Corporation
PhoneExperienceHost.e...		85,748 K	1,58,420 K	12080	Microsoft Phone Link	Microsoft Corporation
CrossDeviceService.exe	< 0.01	49,800 K	1,61,556 K	10864	Microsoft Cross Device Servi...	Microsoft Corporation
RuntimeBroker.exe		2,696 K	14,508 K	12504	Runtime Broker	Microsoft Corporation
TextInputHost.exe	< 0.01	1,25,212 K	1,52,456 K	11632		Microsoft Corporation

At the bottom, the status bar shows: CPU Usage: 19.62%, Commit Charge: 73.15%, Processes: 292, Physical Usage: 74.19%.

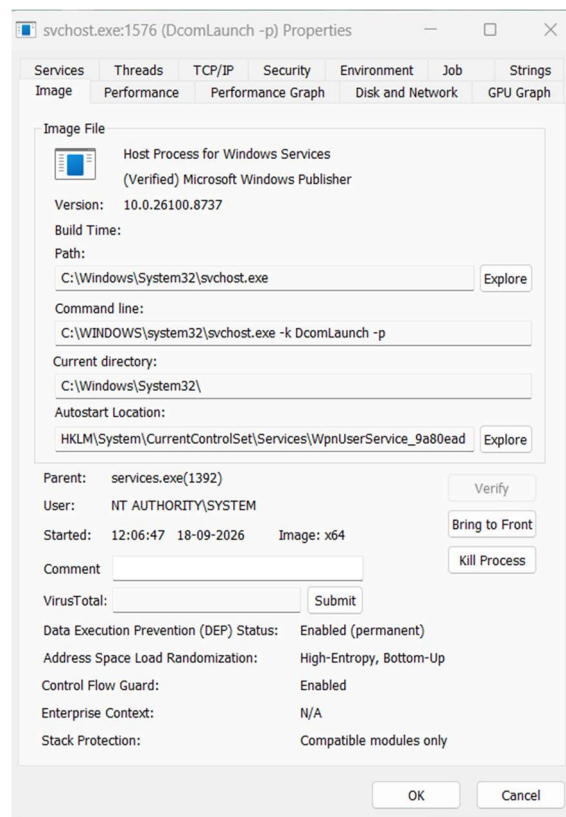
Step 4 – Inspect Process Properties

The `svchost.exe` process was selected and its Properties window was opened.

The Image tab provided the following information:

- Process name: `svchost.exe`
- PID: `1944`
- Description: Host Process for Windows Services
- Path: `C:\Windows\System32\svchost.exe`
- User: `NT AUTHORITY\SYSTEM` ☐ Company: Microsoft Corporation

The executable path was inspected to verify that the process was running from the Windows System32 directory.

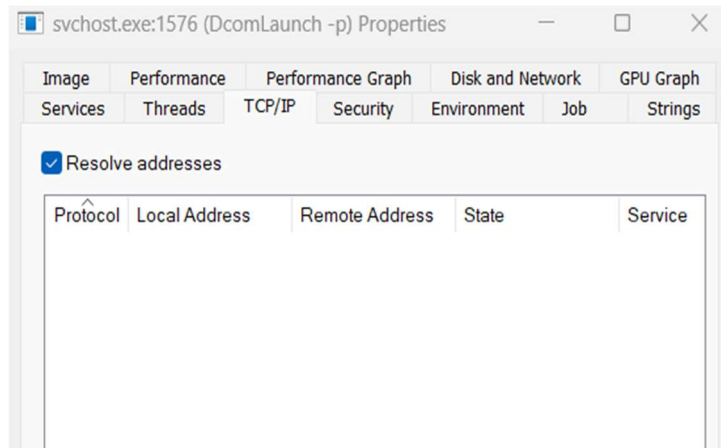


Step 5 – Inspect TCP/IP Activity

The TCP/IP tab of the `svchost.exe` Properties window was opened.

The Local Address, Remote Address, Protocol, State and Service information were inspected.

No TCP/IP connections were displayed for the selected process at the time of inspection.



Step 6 – Examine CPU and Memory Usage

The CPU, Private Bytes and Working Set values were examined for the selected `svchost.exe` process.

The process did not show unusually high CPU usage during the observation.

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
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Step 7 – Online Process Verification

The process name `svchost.exe` was searched online to understand its purpose.

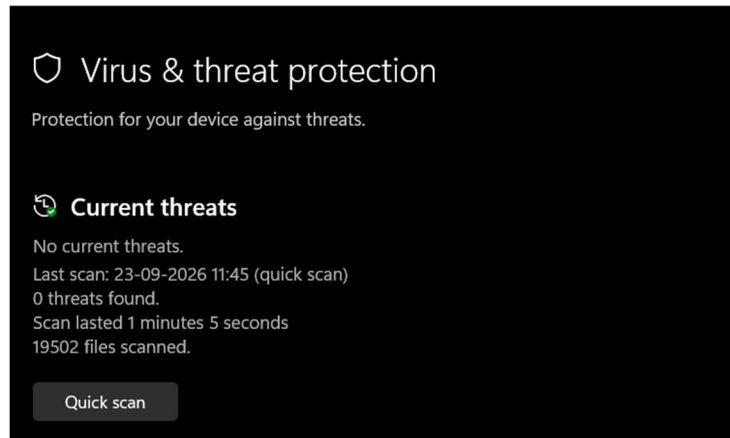
The search results described `svchost.exe` as a Windows system process used to host and manage Windows services.

Step 8 – Antivirus Check

Windows Security was opened and the Virus & threat protection section was examined.

The available scan result showed:

- No current threats
- 0 threats found
- Quick scan completed successfully



Result

Process Explorer was successfully used to examine running Windows processes.

The `svchost.exe` process was investigated by checking its:

- Process ID
- CPU usage
- Memory usage
- Description
- Company name
- Executable path
- TCP/IP activity

The investigation showed that the selected `svchost.exe` process was located in the Windows System32 directory and was associated with Microsoft Corporation. No TCP/IP connections were displayed for the selected process during the inspection.

Conclusion

Process Explorer provides detailed information about running Windows processes and their resource usage. It can be used in digital forensics and system investigation to examine process properties, identify unusual resource usage, inspect network activity, and investigate potentially suspicious processes.